A Hybrid Platform for In-Car Infotainment Systems

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A hybrid operating system approach, with Windows running in 'padded cells' on top of a secure, real-time operating system - all on a single automotive-grade PC - can address the challenging requirements of next-generation infotainment systems.

Demand for more advanced infotainment systems is growing rapidly. In addition to theatre-quality audio and video and GPS navigation, wireless networking and other office technologies are making their way into the car. Despite this increasing complexity, passenger expectations for 'instant on' and high availability remain. At the same time, automobile systems designers must always struggle to keep cost, weight, power, and component size to a minimum. A hybrid operating system approach, with Windows running in 'padded cells' on top of a secure, real-time operating system - all on a single automotive-grade PC - can address the challenging requirements of these next-generation infotainment systems.

The Future of In-car Infotainment

The BMW "Individual" studio is where BMW's most luxurious automobiles are showcased. Climbing into the back of a BMW 745, two video screens, embedded in the front seats, are visible. On the screen is a slick looking “skin” with buttons for a variety of entertainment options. One click turns on the radio and sound of amazing quality resounded. Another button brings up a Windows desktop with icons for Outlook, Word, Internet Explorer, and other staples of the familiar PC environment. In a storage compartment in the back seat a wireless keyboard with integrated mouse is pulled out. A feeling, almost like being at the office. With wireless access to the Internet, it was.

This description is a view into the future of in-car infotainment. In addition to the obvious expectation of audio, video, e.g. DVD, and GPS navigation capability that is common in today’s high end systems, Internet access and Office capabilities are soon to become more common and expected by consumers. However, the automobile environment provides some specific challenges with which Windows is unable to cope.

Problems with Windows-in-the-Car

Although it is known that Windows takes tens of seconds up to minutes to boot, automobile passengers have an expectation for instant-on access to the radio, the GPS navigation system, and other portions of the more traditional entertainment and telematics components.

Furthermore, consumers pay top dollar for upgraded audio systems and expect a concert performance every time they get in the car. Because Windows lacks real-time behaviour, sound playback can sometimes be interrupted, with audible clicks and pops, when the system is juggling other CPU-intensive operations such as loading a large Office document. Windows employs a fairness-based scheduler that performs well for point and click office applications but cannot ensure the flawless audio/visual experience that the car passenger expects.

Finally, although computer users expect Windows to crash occasionally, automobile passengers expect the radio and other traditional components to never fail. In fact, a failure in one of these components is liable to cause a visit to the repair shop. Even worse, a severe design flaw in one of these systems may result in a recall that wipes out the profit on an entire vintage of cars. Exacerbating the reliability problem is a new generation of security threats: bringing Internet into the car exposes it to all the viruses and worms that target networked Windows-based computers.

Just recently, a new cyber attack made headlines: “Worm strikes down Windows 2000 systems”. The worm affected the United States Congress, CNN, and the New York Times. This allows the presumption that vehicles might become an aim for worms and viruses because the car doesn’t have a system administrator to enforce security upgrade policies. Even if there was an effective mechanism to ensure timely upgrades to Windows systems in cars, it is commonly known that security vulnerabilities will continue to be found, and there is no guarantee that a flaw can be patched before it is used to perpetrate an attack. The automobile passenger may not complain about office applications going down, but loss of the complete infotainment system will result in irate customers and increased cost for the automobile manufacturer.

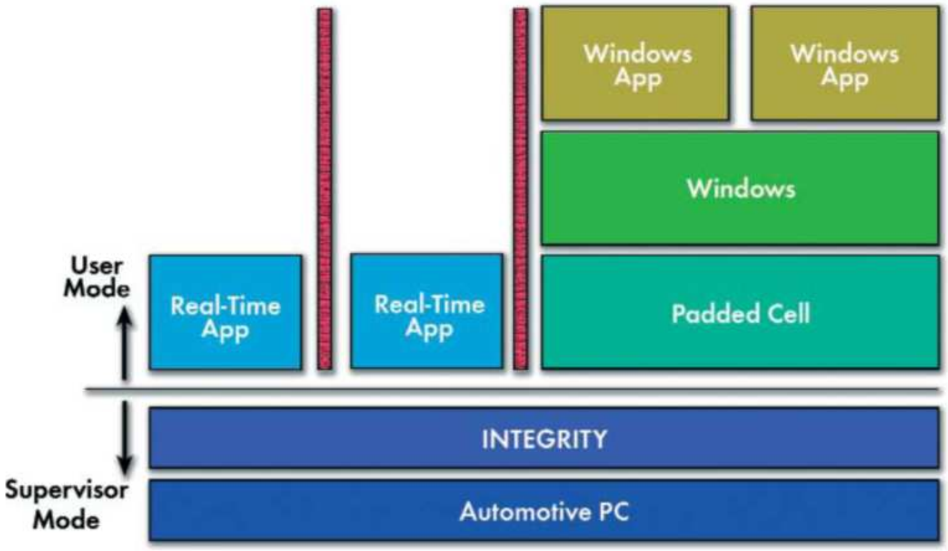

Figure 1: Real-time, critical applications are securely partitioned from Windows and its applications using Padded Cell and Integrity separation kernel technology.

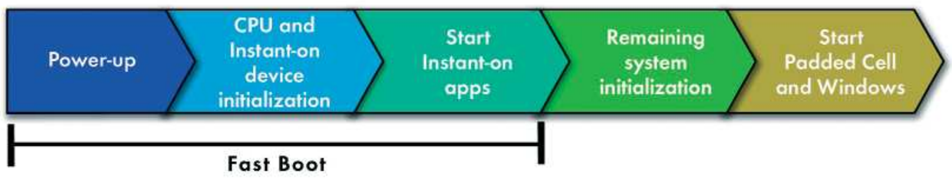

Figure 2: Integrity's fast boot capability ensures that instant-on devices and applications are booted very quickly; Windows initialization can wait.

Solution: A Hybrid Architecture

Clearly, it is not advisable to depend on Windows to control all aspects of the infotainment and telematics systems. Windows cannot boot fast enough, cannot guarantee a real-time, flawless quality of service for audio and other staple functions, and is not reliable and secure enough for automotive platforms.

An obvious solution is to divide the infotainment system onto two independent hardware platforms, placing the high-reliability, real-time components onto a computer running a real-time operating system, and the Windows component on a separate PC. This solution is highly undesirable, however, because of the need to tightly constrain component cost, size, power, and weight within the automobile a solution is required that will run on a single computer.

The Green Hills Software Platform for Automotive is a solution that provides a hybrid secure, real-time operating system coupled with the ability to run Windows on the same computer using secure virtualization technology called Padded Cell. The real-time operating system component is Integrity, the same operating system that controls high reliability systems such as the Joint Strike Fighter and Eurofighter avionics, Boeing 787 flight controls, industrial control systems, life-critical medical devices, and, of course, automotive systems. In addition, the Integrity kernel is in the process of certification to the highest level of security ever achieved (EAL6+) under the Common Criteria, an international standard for security evaluation. In the hybrid architecture, choice of a host operating that has this kind of heritage is critical with respect to meeting automotive-grade reliability expectations, particularly in an Internet-connected environment.

The Green Hills Padded Cell technology provides a high speed virtual machine environment, also known as a Virtual Machine Monitor (VMM). Unlike other commercial virtual machines, Padded Cell runs completely in user-mode along with the guest operating system, e.g. Windows, and its applications. Therefore, none of Windows, its applications, and the virtual machine, itself a complex piece of software, can affect the stability of the real-time, secure components that run directly on top of the Integrity kernel, Figure 1.

Audio and video applications running under control of Integrity are guaranteed to perform flawlessly. Because Integrity is optimized for the extremely fast boot times required by automotive systems, instant-on requirements are met, Figure 2.

Another key aspect of Padded Cell technology is the ability to execute unmodified Windows binaries on existing automotive-grade PC hardware. Thus, infotainment systems developers can take advantage of this architecture without the hassles of maintaining custom Windows kernels and drivers or specialized hardware.

Challenge: Performance

If user-mode execution was the only requirement of hybrid architecture, Green Hills could resort to any number of simulators, such as the open source Bochs product which provides a faithful but slow emulation of the x86/PC architecture. In order to achieve acceptable performance, the virtual machine cannot naively simulate all aspects of the PC architecture.

The mechanism employed by Padded Cell for x86 takes advantage of the fact that much of the code running within the guest environment consists of vanilla user-mode instructions that can be safely executed directly. The virtual machine only needs to intercept code that attempts to execute privileged operations such as memory mapping hardware manipulation. Since the guest operating system executes in user-mode under control of the virtual machine, most of these "dangerous" instructions are automatically trapped by the hardware. When such an instruction traps, the real-time kernel suspends the guest operating system and rapidly notifies the user-mode virtual machine which then performs whatever emulation is necessary to make the guest operating system think that the dangerous instruction was directly executed, Figure 3.

Dangerous instructions that do not cause processor traps are a little more challenging. The virtual machine must detect these instructions before they are executed and ensure that they are safely emulated. The basic idea is to dynamically re-write the guest operating system code. For example, the virtual machine can replace a dangerous instruction with a trap instruction, thereby guaranteeing that the dangerous code goes through the same path as other privileged instructions that automatically generate exceptions. Under Padded Cell, full range video clips, audio included, can be played on a standard, off-the-shelf PC.

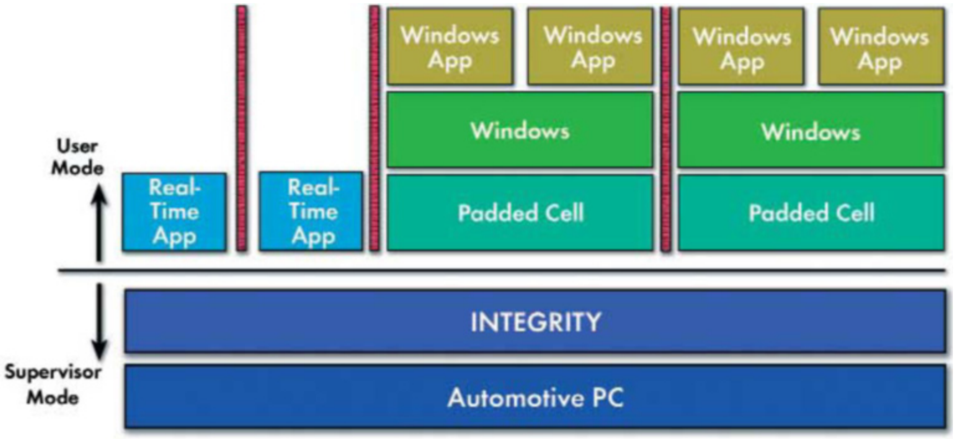

Figure 4: Multiple instances of Padded Cell and Windows enables occupants to enjoy a fully independent infotainment experience despite running on a shared microprocessor.

A Private Virtual PC for Each Occupant

Using Padded Cell technology, multiple instances of Windows, powered by multiple instances of the virtual machine, can run simultaneously on the same computer. In the back seat of the BMW, each passenger has a private video monitor. Without Padded Cell, the passengers would have to share a single copy of Windows. With Padded Cell, one passenger could even reboot Windows without affecting the second passenger's email session, Figure 4. Meanwhile, concert-quality sound running under a native real-time INTEGRITY audio application fills the cabin. Now that is a high-tech infotainment system.

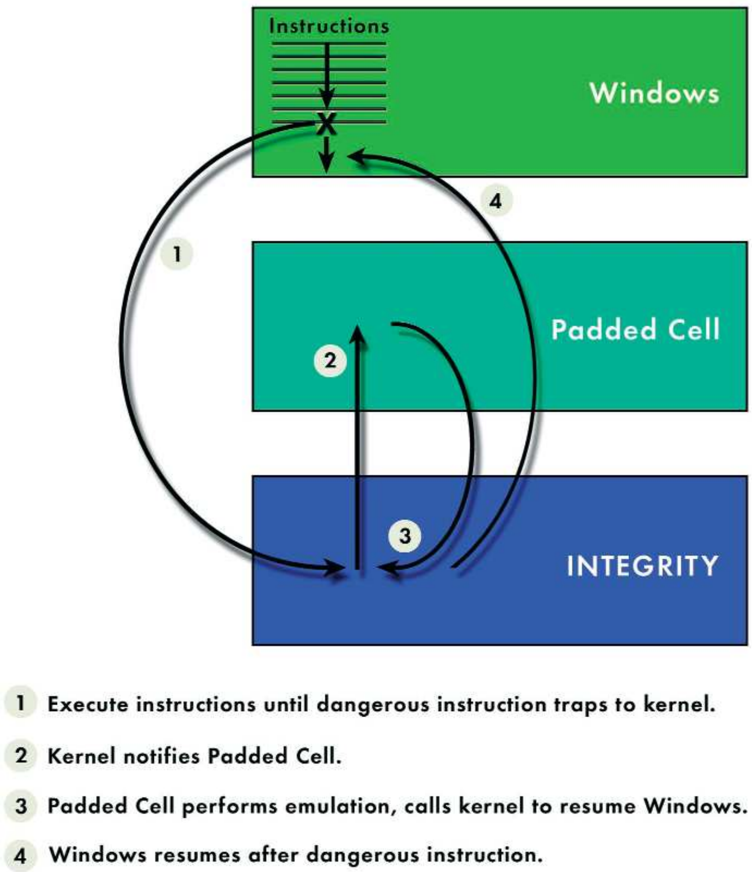

Figure 3: Virtual machine achieves required execution speed by using direct execution coupled with fast context switch and exception handling enabled by a real-time microkernel.